A positive-degree central polynomial below the claimed n^2 bound

Submission for TLMC problem 00000002761

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Scope and conclusion. The source asserts that the least degree of a central polynomial for M_n is n^2. A central polynomial is a nonidentity whose every matrix evaluation is central. As usual, constants are excluded by requiring positive degree. We construct an explicit homogeneous polynomial of degree 8 in the free associative algebra on six variables over $\mathbb{Q}$. Every evaluation in $M_3(\mathbb{Q})$ is scalar, and one evaluation is $2I_3$. Thus the actual least positive degree, with any finite number of variables allowed, is at most $8 < 3^2$. This disproves the degree clause. No conclusion about the source's unspecified moduli space or generation statement is needed. The existence of degree-eight central polynomials for M_3 is also recorded in the literature [1, 2]; the construction and verification below are self-contained.

An explicit weak identity. Put

$$B(X; A, B, C) = \sum_e c_e X^{e_0} A X^{e_1} B X^{e_2} C X^{e_3},$$

where the twenty nonzero coefficients are given by this table. Every listed exponent tuple has sum three.

e	c_e	e	c_e
$(3,0,0,0)$	1	$(0,3,0,0)$	-1
$(2,1,0,0)$	1	$(0,2,1,0)$	-1
$(2,0,1,0)$	1	$(0,2,0,1)$	-1
$(2,0,0,1)$	1	$(0,1,2,0)$	1
$(1,2,0,0)$	-1	$(0,1,1,1)$	-1
$(1,1,1,0)$	1	$(0,1,0,2)$	-1
$(1,1,0,1)$	-1	$(0,0,3,0)$	1
$(1,0,2,0)$	1	$(0,0,2,1)$	1
$(1,0,1,1)$	1	$(0,0,1,2)$	-1
$(1,0,0,2)$	-1	$(0,0,0,3)$	-1

For any expression $F(A, B, C)$, define its alternation by

$$\mathrm{Alt}_{A,B,C} F = F(A,B,C) - F(A,C,B) - F(B,A,C) + F(B,C,A) + F(C,A,B) - F(C,B,A).$$

Let $W(X; A, B, C) = \mathrm{Alt}_{A,B,C} B(X; A, B, C)$. The weak identity needed here is

$$\operatorname{tr} X = 0 \quad \Longrightarrow \quad W(X; A, B, C) = 0 \qquad (X, A, B, C \in M_3(\mathbb{Q})). \tag{1}$$

Here is an entirely finite verification of (1), used also in Lean. Write the bottom-right entry of X as $-X_{00} - X_{11}$, leaving eight independent rational entries. The expression W is alternating and trilinear in A, B, C. For its $(0,0)$ entry, it therefore suffices to use the $\binom{9}{3} = 84$ ordered triples of distinct matrix units. For arbitrary matrices $P_0, \ldots, P_3$, the elementary formula

$$(P_0 E_{ij} P_1 E_{k\ell} P_2 E_{mn} P_3)_{rs} = (P_0)_{ri} (P_1)_{jk} (P_2)_{\ell m} (P_3)_{ns}$$

reduces every case to a polynomial identity in the eight entries. The Lean file proves each resulting identity by rational ring arithmetic, then extends from the actual matrix-unit basis using a multilinear map. This gives $W(X;A,B,C)_{00}=0$ for all A,B,C.

The polynomial commutes with simultaneous conjugation, and trace is conjugation invariant. Consequently every conjugate of $W(X;A,B,C)$ has zero $(0,0)$ entry. Conjugation by permutation matrices first gives all diagonal entries zero. After permuting index j to zero, conjugation by $I+E_{0k}$ with inverse $I-E_{0k}$ gives $T_{jj}+T_{ij}=0$ for each $i\neq j$. Hence every off-diagonal entry is zero as well. This proves (1) without a diagonalizability or density assumption.

From a weak identity to a central polynomial. In each of the twenty block words, mark in turn one of its three occurrences of X. This gives sixty signed decompositions

$$c\,LXR, \qquad c\in\{1,-1\},$$

where L and R retain their order and contain the other letters. Replace the two unmarked occurrences of X by Y. Define

$$D_Y(X;A,B,C)=\mathrm{Alt}_{A,B,C}\sum c\,L(Y)XR(Y),$$
$$G(Y;A,B,C,Z)=\mathrm{Alt}_{A,B,C}\sum c\,R(Y)ZL(Y).$$

These finite formulas define actual noncommutative polynomials, with the sixty terms specified uniquely by the table and the marking rule. Since W is cubic in X, direct expansion gives

$$W(Y+X)-W(Y-X)-2W(X)=2D_Y(X).$$

If X and Y have trace zero, (1) makes the left side zero; over $\mathbb{Q}$ we obtain $D_Y(X)=0$. Cyclicity of trace gives, term by term,

$$\mathrm{tr}(G(Y;A,B,C,Z)X)=\mathrm{tr}(ZD_Y(X;A,B,C))=0 \quad \text{whenever } \mathrm{tr}\,X=0.$$

The annihilator of the trace-zero matrices for this pairing consists exactly of scalar matrices: testing against E_{ji} with $i\neq j$ kills off-diagonal entries, and testing against $E_{ii}-E_{00}$ equates diagonal entries. Therefore $G(Y;A,B,C,Z)$ is scalar whenever $\mathrm{tr}\,Y=0$.

Now define the six-variable free polynomial

$$P(U,V,A,B,C,Z)=G(UV-VU;A,B,C,Z).$$

Every commutator has trace zero, so every evaluation of P in $M_3(\mathbb{Q})$ is scalar. Each marked word contains two copies of Y, the three letters A,B,C, and one Z. Substituting $Y=UV-VU$ therefore makes all its words homogeneous of degree $2+2+1+1+1+1=8$.

A nonzero evaluation and the actual minimum. Use the following matrix units, with indices $0,1,2$:

$$U=E_{00}, \qquad V=E_{01}+E_{02}-E_{20},$$
$$A=E_{22}, \qquad B=E_{10}, \qquad C=E_{20}, \qquad Z=E_{21}.$$

The commutator is $UV-VU=E_{01}+E_{02}+E_{20}$. Direct substitution gives

$$P(U,V,A,B,C,Z)=2I_3.$$

Thus P is not a polynomial identity and is not zero. Its support is nonempty and every support word has length eight, so its degree, defined as the maximum support-word length, is exactly eight; its constant coefficient is zero.

For clarity the formal minimum is an attained minimum, not an assigned number. Let $\mathcal{D}$ consist of the degrees of all positive-degree central polynomials over $\mathbb{Q}$, allowing any finite number of variables. The witness proves $8\in\mathcal{D}$, so $\mathcal{D}$ is nonempty. The natural-number least element $d_{\min}$ exists and satisfies

$$d_{\min}\leq 8<9=3^2.$$

We need not, and do not, prove here that $d_{\min}=8$.

Formal objects and verification. The proof in `lean/Main.lean` uses actual 3×3 rational matrices, the standard matrix-unit basis, and the native free algebra $\mathbb{Q}\langle \mathrm{Fin}\, 6 \rangle$. Its canonical isomorphism with the monoid algebra of the free monoid supplies the coefficient support; the evaluation bridge identifies the native universal evaluation map with the polynomial formula above. The concluding declarations are `native_polynomial_witness` and `counterexample` in namespace `TLMC2761`.

The original bilingual source is retained in `SOURCE.md`. The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

Its evidence includes every namespace declaration's axiom audit, the exact build and fresh-kernel checks, rendered-PDF review, and file hashes. The formula was independently derived and is proved directly; we do not identify it with an unread historical formula.

References

[1] V. Drensky and B. Kostadinov, *Central polynomials of minimal degree for matrices*, introduction, arXiv:2601.07750v1.

[2] V. Drensky, *Weak polynomial identities and their applications*, Communications in Mathematics 29 (2021), no. 2, 291–324, especially p. 303, DOI 10.2478/cm-2021-0022, author's survey.